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From top to bottom: gridded human population estimates in data-poor situations

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Summary

Where disease risks are heterogeneous across population groups or space, or dependent on transmission between individuals, spatial data on population distributions – human, livestock and wildlife – are required to estimate infectious disease risks, burdens and dynamics. As a result, large-scale, spatially explicit, high-resolution human population data are being increasingly used in a wide range of animal- and public-health planning and policy development scenarios. Official census data, aggregated by administrative unit, provide the only complete enumeration of a country's population. While census data from developed countries are generally up-to-date and of high quality, in resource-poor settings they are often incomplete, out of date, or only available at the country or province level. The challenges associated with producing accurate population estimates in regions that lack high-quality census data have led to the development of census-independent approaches to small-area population estimations. Known as bottom-up models, as opposed to the census-based top-down approaches, these methods combine microcensus survey data with ancillary data to provide spatially disaggregated population estimates in the absence of national census data. This review highlights the need for high-resolution gridded population data, discusses problems associated with using census data as top-down model inputs, and explores census-independent, or bottom-up, methods of producing spatially explicit, high-resolution gridded population data, together with their advantages.

Keywords

Bottom-up models – Census independent – Gridded population data – Microcensus surveys.

Introduction

Large-scale, spatially explicit, high-resolution human population data are increasingly used in a wide range of animal- and public-health planning and policy development scenarios. Examples include:

- defining disease risk [1, 2];
- estimating risk of infection and size of the population at risk [3, 4, 5, 6];
- estimating disease burdens and identifying target populations [6, 7];
- setting targets and measuring progress [8];
- informing management and policy decisions [3, 9];
- siting health facilities [10];
- assessing the impact of natural disasters [11, 12, 13];
- investigating disease dynamics [14, 15];
- forecasting disease risk distribution under changing climate and land use conditions [16];
- evaluating the epidemiological, environmental and socio-economic impacts of the human and livestock sectors [17];
- acting as the population-at-risk denominator for epidemiological studies.

Additionally, gridded population data are a strategic resource supporting the monitoring and implementation of international frameworks such as the Sustainable Development Goals (SDGs) [18] and the Sendai Framework for Disaster Risk Reduction [19]. In particular, the SDGs use population data as a denominator when calculating some of their 231 unique indicators (e.g. the proportion of the population living below the national poverty line, or the proportion of men, women and children of all ages living in poverty in all its dimensions according to national definitions) [20]. Ideally, to be of greatest use in spatial and modelling applications, these large-scale, spatially explicit, high-resolution human population data should be 'up-to-date, sufficiently

detailed, accurate, consistent, cost-effective (i.e. sustainable), transparent (i.e. using clear methods), and accessible to all' [21].

Sourcing detailed population data: from census to grids

Official census data provide the only complete record of a country's population [22]. For reasons of confidentiality, the final released data are aggregated by administrative unit, either in tabular format or as choropleth maps (vector maps comprising polygonal representations of population totals per administrative unit). However, aggregated data, and choropleth maps in particular, have several limitations, including loss of detail, masking of within-area spatial heterogeneity, and the modifiable area unit problem [23] – the latter referring to the statistical bias introduced through data aggregation, with both shape and scale of the aggregating units influencing the resulting summary values [24]. As such, aggregated data are unable to describe accurately the dynamic and continuous structure of human populations [25]. Moreover, prior to 1990 data were only aggregated by country and major sub-divisions (provinces or states), and although these data are now more readily available by district, subdistrict and enumeration areas [22], the size of national and subnational administrative units can vary considerably between countries, making accurate spatiotemporal comparisons between them challenging [22].

Instead, for data that are to be used for spatial analyses or modelling, it is preferable and more accurate to provide population estimates as gridded (i.e. per square kilometre [km^2]), rather than aggregated, data. Although few national statistical offices produce population data in this format [22], a range of techniques have been developed for converting the commonly available aggregated data to population grids [26] using dasymetric disaggregation, or top-down modelling [27]. Dasymetric disaggregation uses ancillary data in the form of covariates that are known to be associated with human distribution (e.g. climate and landcover) to redistribute the aggregated data in order to present a more realistic population distribution [27]. As well as providing a more accurate representation of the heterogenous nature of population distributions, this gridding process can remove uncertainties associated with the originally reported administrative units and provide data that can be integrated with other spatial features that are regularly portrayed as grids (e.g. climate data) [28].

The main gridded population data sets have seen a considerable increase in resolution concomitant with an increase in computational processing power and granularity of ancillary data [29] and advances in classification of populated land from remote sensing data [30]. Currently, many top-down gridded data sets use a 30-second arc resolution (~1 km at the equator), although some data sets, including WorldPop (~100 metres [m] at the equator) and Environmental Systems Research Institute's World Population Estimate (150 m), use a much higher resolution. In general, using the highest-resolution data available has been shown to improve accuracy and prediction precision [31] but must also provide a meaningful analytical unit for the research [27].

Top-down gridded population estimates

The distribution and density of human populations, both global and local, has always been influenced by a range of geographical factors. A study of 32 low- and middle-income countries found that, despite a few minor regional differences, these drivers of population distribution are remarkably consistent [29], with covariates relating to settlement, ecology, climate and topography explaining most of the small-scale variation in population distribution [29]. These covariates are also referred to as ancillary data, and dasymetric mapping techniques exploit the known relationships between these covariates and population distribution to disaggregate census-based data and redistribute the aggregated data more realistically [27]. Ancillary data used frequently in such top-down models include landcover [32, 33, 34, 35, 36, 37], nighttime lights [33, 38, 39, 40, 41, 42], road networks and buffers [32, 39, 43], waterbodies [43], points of interest [40], urban areas [21, 34, 43] and settlement extent [36]. Novel ancillary data include the use of geotweet density (i.e. density of geotagged Twitter posts) [44, 45].

Top-down modelling approaches have been divided into unmodelled (areal weighting), lightly modelled (areal weighting with ancillary data) and highly modelled (statistical and machine learning modelling) methods depending on the modelling techniques and ancillary data used [27]. Of the open access data sets available, Gridded Population of the World [46] is considered unmodelled; lightly modelled open access data sets include Global Rural-Urban Mapping Project [47] and Global Human Settlement Layer Population [48, 49]; and highly modelled data sets include LandScan Global Population Dataset [50] and WorldPop [51]. Understanding the differences between these approaches and data sets is fundamental to assessing the fitness of any single one for a specific use, and the various approaches and resulting data sets have been extensively reviewed by Balk *et al.* [22], Leyk *et al.* [27] and Wu *et al.* [52].

However, although much research in this area has focused on determining the best statistical methods for disaggregating census data and identifying the ancillary data that most accurately portray population distribution [27], the main challenge for top-down modelling remains the accuracy of the input data as top-down gridded population data are only as good as the census data on which they are based [30].

Limitations of census data as inputs for dasymetric models

National censuses are usually held every ten years [53] and are 'among the most complex and massive peacetime exercises a nation undertakes' [54]. Although scheduled to take place every ten years, census years are not synchronised between countries [22] and therefore any between-country comparisons need to be harmonised to a common base-year [22]. In addition, high-resolution administrative boundaries often change between census years, making spatial comparisons over time challenging [32]. While census data from developed countries are generally up-to-date and of high quality, in resource-poor settings – which include large parts of the low- and middle-income countries – current and accurate census data are frequently unavailable because they are incomplete, out-of-date, or only available at the country or province level [30]. Furthermore, any available data may quickly become outdated during the intercensal period owing to rapid, non-uniform population growth as a result of migration, conflict-related displacement and high urbanisation rates [30], especially if the intercensal period stretches for longer than the purported ten years [30].

The situation is further aggravated when estimates are projected forward, as many developing countries display differential growth rates among census units rather than the constant growth rates used in official dasymetric mapping projections [55]. For example, as a result of substantial development, the Kano metropolitan area of Nigeria showed an annual growth rate of 2.02% for 2000–2006, but this then increased to 4.37% per annum for the years 2006 to 2014, rather than maintaining the constant growth rate officially used for projections [56]. As a result, while top-down approaches to population estimates are suitable for regions with reliable census data, they are limited when census data are inadequate or a poor reflection of the true population, as any inadequacies in the census data will cascade down to the resulting population grids. The challenges associated with producing accurate population estimates in data-poor regions are one of the main reasons behind the development of census-independent approaches to small-area population estimations [30]. Also known as bottom-up methods, they have been highlighted by the United Nations as having a key supporting role to play in estimating population density and distribution in data-poor countries [54].

Bottom-up gridded population estimates

Unlike top-down models that use dasymetric disaggregation of aggregated census data, which by definition have been collected for entire regions, the objective of bottom-up models is to predict population density and distribution for a large region using data collected from only a fraction of that region [30]. Instead of large, comprehensive census data, bottom-up approaches use microcensus surveys [57] – complete counts or enumerations of populations within small, well-defined areas of interest [56]. Similar to top-down methods, statistical or machine learning models then combine the microcensus survey data with relevant ancillary data to predict population numbers in unsampled regions [30]. Moreover, machine-learning techniques and increased computational power have combined with high-resolution satellite imagery to allow human-built structures and settlements to be identified, and with this knowledge, population data can be more accurately allocated to residential areas at increasingly finer scales, rather than simply disaggregating data to an entire region [30].

Early bottom-up efforts to disaggregate microcensus data used ancillary variables similar to those used in top-down dasymetric approaches, such as nighttime lights [58], or vegetation indices and temperature [59]. However, as the focus of bottom-up approaches has moved towards estimating total population of urban areas with high spatial precision [56], there has been a move towards using ancillary data specific to settled areas, which allows for disaggregation of these urban data to provide a detailed distribution of residents per unit of settlement area. For example, Weber *et al.* [56] first used the Settlement Mapping Tool (SMT) platform – a machine learning system for extracting settlement areas from high-resolution imagery that uses a support vector machine classifier to identify settled and non-settled regions [60] – to divide their settlement area into developed (1.46%) and undeveloped (98.40%) land, before disaggregating the population throughout the 1.46% of developed land.

Obtaining building footprint data

Early approaches to bottom-up models focused on identifying and delineating the building footprint [56]. Using high-resolution settlement imagery, rooftops were manually delineated [61], but manual methods have since developed into automated, scaled-up approaches of settlement mapping [56] using, for example, the SMT platform [60]. In addition to creating individual building footprint data sets from high-resolution satellite imagery for specific areas of study, there is an increasing range of building footprint data sets available. Sources of volunteered geographic information, such as OpenStreetMap (https://www.openstreetmap.org), have added additional settlement data, providing details on building footprints [56] to aid humanitarian efforts. For example, during the 2014 West Africa Ebola crisis, when faced with only a few rudimentary topographical maps of Guinea but no useful maps upon which to base control and surveillance efforts, Médecins Sans Frontières personnel enlisted the help of the Humanitarian OpenStreetMap Team (https://www.hotosm.org) to map Guéckédou, the main city in Guinea affected by the outbreak. Within 20 hours of receiving the request, online volunteers had mapped three cities in Guinea based on satellite imagery of the area, populating them with over 100,000 buildings – information that proved

crucial for door-to-door canvassing of inhabitants and mapping the spread of disease [62].

These efforts have since escalated and an increasing range of open access data sets of building footprints exists, including Microsoft's Bing Maps (https://www.microsoft.com/en-us/maps/building-footprints), which provide building footprint data for North and South America, Australia and four African countries (Uganda, Tanzania, Nigeria and Kenya); and the Ordnance Survey OpenMap – Local (https://osdatahub.os.uk/downloads/open/OpenMapLocal), which provides building footprints for all of Great Britain. The building footprints of individual cities such as New York (https://data.cityofnewyork.us/Housing-Development/Building-Footprints/nqwf-w8eh), Washington, DC (https://opendata.dc.gov/datasets/DCGIS::building-footprints/about) and Chicago (https://data.cityofchicago.org/Buildings/Building-Footprints-current-/hz9b-7nh8) are also available as open access data.

Improving model accuracy with building characteristics

Although the existence and extent of buildings are suitable for delineating settled from unsettled areas, the information is insufficient for good-quality dasymetric disaggregation within settled areas, as Weber *et al.* [56] note that residential population density is 'the multiplicative product of several variables, including the spacing of buildings, building height, and the number of residents per floor area within households' [56]. Differentiating between settled and unsettled areas is therefore an important first step in bottom-up modelling [56]. However, model performance can be greatly improved through additional layers of information [56], including types of settlements (residential or non-residential) [63] or building characteristics [64]. Including residential type has been shown to reduce the model's prediction intervals [56], and at fine spatial scales, areas with homogenous residential types also provide narrower prediction intervals [56].

The ancillary data used in the disaggregation of residential populations have therefore expanded to include a range of building characteristics, and studies have used rooftop areas [61]; building footprints and height [65]; building density, height and type [66]; and function [30] to disaggregate these data. Moreover, Weber *et al.* [56] suggest using a combination of ancillary factors, rather than one or two, to capture the multiple cultural and socio-economic factors that influence population density within settled areas. In this way, the predictive accuracy of disaggregated residential population data is greatly improved [56].

Building on this, the R package *foot* allows a range of morphology metrics to be extracted from geospatial building data sets – either for individual building footprints, summarised for different area levels, or as gridded data sets [67], including area, perimeter, roundness, compactness, angle of rotation (measure of structure rotation) and nearest neighbour distance (buildings clustered or dispersed). Future developments of the *foot* package will include the potential for users to define their own metrics [67]. The developers of the *foot* package demonstrate its functionality using these metrics to derive a settlement typology map for Great Britain, easily differentiating between urban, suburban and rural areas, with additional layers of classifications within each area [67].

Where no population data are available, either from censuses or microcensus surveys, WorldPop's peanutButter web application and R package [68] allow gridded population estimates to be produced using a researcher's own estimates of people per building. Using the 'peanut butter' method, the researcher's estimates of average household sizes for different settlement types (e.g. urban, peri-urban, rural) are spread evenly (like peanut butter) across buildings in each settlement type. The building footprints used in the application are high-resolution maps of building footprints based on recent satellite imagery [69].

Problems classifying settlements

Determining the building area fraction requires settlement mapping methods that precisely outline buildings. Despite using very high-resolution satellite imagery, the accurate detection of buildings in rural areas is not always possible as small buildings may be partly obscured by clouds, or by smoke from slash and burn agriculture [56]. This can lead to population underpredictions in rural areas [70]. In addition, platforms such as SMT [60] expect settlements to be large, and to contain buildings made up of multiple straight lines and built from high reflectance materials, such as concrete, glass or steel. Misclassification can therefore occur with very small or rural settlements, or when the buildings are round huts (no straight lines) with thatched rooftops (minimal reflectance), leading to errors of omission [56]. Alternately, errors of commission can arise when linear features such as riverbanks or roads are misclassified as buildings [56].

Advantages of bottom-up over top-down disaggregation

While top-down methods are suitable for large-scale estimates and regions with good data, bottom-up or census-independent methods are potentially useful for data-poor regions [56]. With a focus on settled areas, bottom-up population estimates usually have very high spatial precision, with resolutions of 100 m [70], 10 m [71] or even 8 m [30]. By modelling settlement area directly and for individual years, it is possible to highlight areas undergoing rapid urbanisation by identifying changes to a settlement's building footprint, which allows appropriately variable growth rates to be used in future population predictions rather than the conventional

constant growth rates [56]. As microcensus data are usually collected at the individual rather than the household level [57], population estimates arising from these data can be stratified by a range of key demographics including age and sex [56, 70], and it is possible to quantify the uncertainty inherent in estimating populations from a microcensus survey through the use of associated uncertainty measures for the area units of interest using such measures as prediction intervals [56], bootstrap statistical methods [61] or Bayesian posterior probabilities [70, 71].

Top-down estimates are generally based on large-scale algorithms applied consistently across the regions of interest (global [72], regional [37, 73] or national [74]) and are used to serve broad, general needs (e.g. disease control, land planning, climate change). Bottom-up approaches, with their focus on mapping settled areas with very high resolution (10 m), lend themselves to applications of smaller spatial scales. Although these two approaches are currently distinct entities, future research will likely combine the best of both. At the very least, bottom-up estimates can be used as a source of local ground-truthing to refine national census-based estimates.

Conclusions

Although not a replacement for the census-dependent, top-down population grids, bottom-up methods are a valuable alternative, particularly for countries with limited or outdated census data. Given the extremely high resolution at which some of these data sets are available (8 to 10 m), together with their focus on residential areas and ability to estimate true growth rates, these data sets provide a strategic resource for public-health planning and policy development scenarios, and to support the monitoring and implementation of international frameworks.

Partir du bas plutôt que du haut : les estimations maillées des populations humaines dans les situations où les données sont rares

K.B. Stevens

Résumé

Dans les contextes où les risques de maladie sont hétérogènes et dépendent du groupe de population ou du territoire dont il s'agit, ou des probabilités de transmission entre individus, il est nécessaire de disposer de données spatiales sur la distribution des populations (couvrant la population humaine et les populations d'animaux d'élevage et sauvages) afin d'être en mesure d'évaluer les risques de maladie infectieuse, de calculer le fardeau qu'elles représentent et de mettre en lumière les dynamiques à l'œuvre. En conséquence, il est fait de plus en plus souvent appel à des données spatialement explicites, à grande échelle et à haute résolution pour construire les scénarios utilisés à des fins de planification et d'élaboration des politiques de santé animale et de santé publique. Les données officielles de recensement agrégées par unité administrative constituent la seule énumération complète de la population d'un pays. Si dans les pays développés ces données de recensement sont généralement actualisées et de bonne qualité, dans les configurations moins dotées en ressources elles sont souvent incomplètes, obsolètes ou n'existent qu'à l'échelle nationale ou provinciale. Les difficultés rencontrées pour produire des estimations suffisamment exactes dans les régions dépourvues de données de recensement de bonne qualité ont conduit à élaborer des méthodes visant à estimer la population de territoires limités, sans passer par le recensement. Ces modèles, qualifiés d'« ascendants » par opposition aux modèles de recensement « descendants », associent aux données issues d'opérations de micro-recensement un certain nombre de données complémentaires afin de fournir des estimations de population ventilées par territoires, en l'absence de données nationales de recensement. Dans cet article, l'auteure souligne l'importance de disposer de données maillées de population à haute résolution ; après avoir examiné les problèmes associés à l'utilisation des résultats des modèles descendants, elle décrit les méthodes ascendantes non basées sur le recensement et leur capacité à fournir des données maillées de population spatialement explicites et à haute résolution. Elle conclut sur les avantages de ces dernières méthodes.

Mots-clés

Données maillées de population – Méthode non fondée sur le recensement – Micro-recensement – Modèles ascendants.

No desde arriba, sino desde abajo: el cálculo con cuadrículas de población humana cuando escasean los datos

K.B. Stevens

Resumen

En circunstancias en que el riesgo de enfermedad varía según el grupo de población o el espacio de que se trate o en que dicho riesgo depende de la transmisión entre individuos, es necesario disponer de datos espaciales sobre la distribución de poblaciones (ya sean humanas, ganaderas o de animales salvajes) para calcular el riesgo y determinar la carga y la dinámica de una enfermedad infecciosa. De ahí que en muy diversas situaciones en las que se elaboran planes o políticas de sanidad animal o salud pública se vengan utilizando, cada vez más, conjuntos de datos a gran escala y alta resolución referidos expresamente a la población humana de un determinado ámbito geográfico. Los datos del censo oficial, agregados por unidad administrativa, ofrecen el único recuento completo de la población de un país. No obstante, si bien los datos censales de países desarrollados suelen estar al día y ser de buena calidad, en condiciones de escasez de recursos esos datos tienden a ser incompletos, estar obsoletos o existir únicamente a nivel de país o de provincia. La dificultad de obtener estimaciones poblacionales exactas en regiones donde no hay datos censales de buena calidad ha llevado a concebir métodos que no dependan del censo para realizar cálculos referidos a la población de pequeños territorios. Estos métodos, llamados modelos «ascendentes», por oposición a los planteamientos «descendentes» basados en el censo, permiten subsanar la falta de datos censales nacionales combinando datos de encuestas microcensales con otros datos complementarios para obtener estimaciones poblacionales desglosadas por espacio geográfico. La autora, tras subrayar la necesidad de disponer de cuadrículas de población de alta resolución, explica los problemas derivados del uso de datos censales como fuente de información en los modelos «descendentes» y expone métodos no dependientes del censo, o «ascendentes», para elaborar cuadrículas de población de alta resolución referidas expresamente a un espacio geográfico, así como las ventajas que ofrecen estos métodos.

Palabras clave

Datos de cuadrículas de población – Encuestas microcensales – Modelos ascendentes – No dependencia del censo.

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